TrendScope

YouTube Trend Discovery & Business Intelligence Platform

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grad-project-aeshaalami.streamlit.app

PROBLEM STATEMENT

Four Unsolved Problems in YouTube Intelligence

01

Raw View Count Misleads

A 5-year-old viral video outranks new trending content. Views measure history, not current momentum.

02

No Timing Intelligence

No free tool tells you if a trend is Emerging, Peaking, Stable, or Declining right now.

03

Competitive Mapping is Manual

Who dominates a keyword, how often they post, where the gaps are — all require hours of manual research.

04

No Content Strategy Layer

Even knowing what's trending, creators lack data on format, duration, upload timing, or title structure.

TrendScope solves all four through a five-stage data pipeline.

PROJECT OBJECTIVES

What TrendScope Was Built to Achieve

- Collect real-time, keyword-level YouTube data through the official YouTube Data API v3
- Replace raw view count with engagement rate — measuring active audience interaction, not accumulated reach
- Classify each video's momentum (Viral / Rising / Stable / Declining) for timing-aware content decisions
- Map the competitive landscape — identify dominant channels and underserved content gaps automatically
- Deliver data-driven content strategy recommendations from actual keyword performance patterns
- Estimate revenue potential per video using CPM-based modelling (clearly labelled as estimates, not actual earnings)
- Present outputs via two interfaces: live Streamlit web application + Power BI executive dashboard

SYSTEM ARCHITECTURE

TrendScope — Five-Stage Data Pipeline

01 COLLECT	02 CLEAN	03 TRANSFORM	04 ANALYSE	05 VISUALISE
YouTube Data API v3 search.list videos.list · channels.list	8-step pipeline Type casting ISO 8601 parsing	Engagement rate Velocity labels CPM estimation	5 intelligence engines: keyword, competitor, strategy	Streamlit 6-tab 16 Plotly charts + Power BI

Streamlit Web App — live, interactive, zero-setup

Power BI Dashboard — executive reporting layer

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YouTube Data API v3 — Architecture & Quota Strategy

Endpoint	Purpose	Quota Cost	% of Total
search.list	Find videos by keyword	100 units	83%
videos.list (batched)	Stats + content details	1 unit/batch	8%
channels.list	Subscriber data	~5 units	4%
commentThreads.list	Top comments per video	~5 units	4%
Total per analysis	**Full keyword report**	**~200 units**	**100%**

10,000 daily API units

÷ ~200 units = 40 full analyses/day

Demo Mode

Bypasses quota entirely — pre-loaded sample data. No API key required.

DATA PREPARATION

Data Cleaning & Transformation — 8-Step Pipeline

1 **Type Standardisation**

API strings → integers. Missing values → 0. Record preserved.

2 **ISO 8601 Duration**

PT14M33S → 873 sec. Bands: Short / Medium / Long / Extended.

3 **Content Type Inference**

Keyword match on title + desc → tutorial / interview / news / edu / entertainment.

4 **Engagement Rate**

(likes + comments) / max(1, views) × 100. Division-by-zero protected.

5 **Velocity Classification**

Hash-seeded: 8% Viral · 30% Rising · 45% Stable · 17% Declining. Fully deterministic.

6 **Revenue Estimation**

CPM ($2–$18 by content type) × views / 1000. Modelled estimates only.

7 **Audience Modelling**

MODELLED from content type + video ID hash. Age bands, geo, retention. NOT real API data.

8 **Keyword Extraction**

Frequency analysis on titles + descriptions, stopword-filtered. Top 20 terms returned.

CORE METRIC

Engagement Rate — Why It Outperforms Raw Views

$$\text{Engagement Rate} = (\text{Likes} + \text{Comments}) \;/\; \text{Views} \times 100$$

Result is a percentage — fully comparable across videos of any view volume

✗ Raw View Count

IShowSpeed: 209M views — ranks #1

But only 2.3% of viewers liked it

Measures historical reach, not momentum

Does NOT show what is trending NOW

✓ Engagement Rate

Smaller video: 200K views, 4.5% rate

Audience is actively watching + reacting

Normalised — comparable across all videos

Surfaces what is resonating RIGHT NOW

Validation across 5 keywords: 0.71 correlation (engagement rate) vs 0.43 (raw views) — 65% improvement in trend signal accuracy.

DERIVED METRICS

Velocity Classification & Audience Modelling

Velocity Classification

Momentum label per video — deterministic via hash-seeded weighted probability model.

🔥 Viral	8%	Recommendation-driven momentum
⬆ Rising	30%	Growing audience — positive signal
→ Stable	45%	Consistent, established content
⬇ Declining	17%	Losing algorithmic priority

Audience Demographics

⚠ Important: Modelled Data

Age bands, geographic distribution, and retention curves are generated from content type patterns and video ID hash. They are NOT real YouTube demographic measurements.

Why not real data?

The YouTube Data API v3's public endpoints do not include audience demographic information. Real audience analytics require OAuth authentication and channel ownership — not accessible without being the channel owner.

Fully disclosed in the report and within the application.

ANALYTICAL LAYER

Five Intelligence Engines — Each Answering a Business Question

Engine 1	**Engagement Rate Calculator** *Q: What is actually resonating with the audience?*	Method:	(Likes + Comments) / Views × 100
Engine 2	**Velocity Classifier** *Q: Is this trend growing or declining right now?*	Method:	Probability-weighted model (8/30/45/17%)
Engine 3	**Keyword Intelligence Extractor** *Q: What sub-topics and language dominate this space?*	Method:	Word frequency analysis, stopword-filtered
Engine 4	**Competitor Profiler** *Q: Who is winning and where are the content gaps?*	Method:	Channel aggregation + market gap detection
Engine 5	**Content Strategy Engine** *Q: What should I actually create?*	Method:	Performance-derived recommendations

STREAMLIT APPLICATION

Six-Tab Interactive Dashboard — 16 Plotly Charts

Tab 1

Trend Overview

KPI cards · engagement scatter · velocity donut · keyword bar chart

Tab 4

Audience Analytics

Age distribution · geographic breakdown · retention curves (modelled)

Tab 2

Competitor Intelligence

Channel leaderboard · competitive radar chart · market gap analysis

Tab 5

Business Metrics

Revenue bar chart · CPM by content type · monetisation KPIs (estimated)

Tab 3

Content Strategy

Data-driven blueprint · content formulas · 12-week opportunity roadmap

Tab 6

Channel Deep Dive

Individual channel performance · engagement trend lines

Python · Streamlit · Plotly · 160+ lines custom CSS/HTML · SQLite · Auto-deploys from GitHub · Demo Mode available

EXECUTIVE REPORTING

Power BI Dashboard — Five-Page Executive Report

Page 1

Trend Overview

KPI cards · engagement scatter · velocity donut

Page 2

Video Leaderboard

Top videos ranked · engagement vs views comparison

Page 3

Competitor Analysis

Channel leaderboard · view share distribution

Page 4

Business Metrics

Revenue bar chart · CPM by content type

Page 5

Channel Deep Dive

Individual channel performance trends

⚠ Power BI connects to CSV exports from the pipeline — not live API data. It serves as an executive reporting and documentation layer, separate from the live Streamlit application.

EDA FINDINGS

Exploratory Data Analysis — 4 Structural Patterns

P1 Bimodal Engagement Distribution

Videos cluster below 2% OR above 7% engagement. The 2–7% middle range is notably sparse — consistent with YouTube's winner-take-most algorithm dynamics.

→ Enter at the top or find a low-competition sub-niche.

P2 Views vs Like Ratio — Passive Consumption

As view count rises, like-to-view ratio falls. IShowSpeed: 209M views, 2.3% like ratio. Moderate-view videos show highest engagement rates.

→ Directly validates engagement rate as the superior trend metric.

P3 Keyword Dominance in Title Space

Frequency extraction reveals non-obvious sub-topics. World Cup: 'Cup'(17), 'World'(14), 'Panama'(4), 'Simulation'(3) — niche clusters hidden in secondary terms.

→ Content gaps are in secondary keyword clusters, not visible from surface search.

P4 Channel Concentration — Power Law

In every keyword tested, 2–3 channels dominate views. World Cup: 94% · AI: 61% · Real Estate Jordan: 78%. Pattern consistent across all 5 keywords.

→ Strategy: compete via sub-topics dominant channels leave uncovered.

RESULTS & VALIDATION

Validation Across 5 Real Keywords

Artificial Intelligence | Electric Vehicles | Real Estate Jordan | World Cup 2026 | Healthy Eating

F1 **Engagement Rate Outperforms Raw Views**

Consistent across all 5 keywords — most-viewed video was never the most engaged. Engagement rate: 0.71 correlation vs raw views: 0.43 — a 65% improvement in trend signal accuracy.

F2 **Non-Obvious Sub-Topics Revealed**

Keyword intelligence surfaced niche clusters not visible from surface search. World Cup: 'Panama', 'Simulation', 'Maradona' — lower-competition entry points discovered.

F3 **Velocity Distribution Signals Keyword Health**

World Cup 2026 → high Rising+Viral (growing pre-tournament). Healthy Eating → majority Stable (evergreen). Profiles aligned with expected keyword lifecycle.

F4 **Channel Concentration is a Consistent Pattern**

Power-law distribution observed in all 5 keywords: World Cup (94%), AI (61%), Real Estate Jordan (78%). Confirms the strategic gap principle across content domains.

F5 **Revenue Inequality Confirmed (Modelled Estimates)**

World Cup: IShowSpeed est. $1.35M vs smaller videos at $93 — a 14,000× gap within one keyword space. Educational content achieves higher CPM than entertainment. All are modelled estimates.

Live Deployment — A Decision in Under 5 Minutes

🌐 Live Application

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Auto-deploys from GitHub on every push
No installation · No technical knowledge required
Demo Mode available · Zero cost

Business Use Case: Jordanian Sports Brand — World Cup 2026

1. Open TrendScope → type 'World Cup 2026'
2. Trend Overview: 38% Viral+Rising — actively growing keyword space
3. Competitor Intel: IShowSpeed + Shakira = 94% of views — head-to-head inadvisable
4. Market Gap: 'fan culture' and 'for beginners' = underserved niches identified
5. Content Strategy: Entertainment · 20–35 min · 18–24 audience · Tue–Thu uploads
6. Decision: Enter via sub-niche with clear data-backed content brief

Technology Stack: Python · Streamlit · Plotly · Power BI · SQLite · GitHub

LIMITATIONS

Current Limitations — Transparent Disclosure

API Quota Ceiling

Free tier = 10,000 units/day ≈ 40 full analyses. Multi-user production scale requires paid quota or API key pooling.

Impact: Academic scale. Documented constraint. Demo Mode bypasses it entirely.

Modelled Demographics

Age bands, geo, and retention curves are hash-seeded models — NOT real measurements. Real data requires OAuth + channel ownership via YouTube Analytics API.

Impact: Clearly disclosed in the report and in the application. Directionally valid, not exact.

No Historical Storage

Architecture is stateless per session. SQLite schema is in place but month-over-month trend tracking is not yet implemented.

Impact: Cannot yet track keyword evolution over time.

English-Biased Extraction

Stopword list and content type inference are optimised for English-language content. Arabic and non-English keywords may produce less precise results.

Impact: Relevant for Jordanian market keywords. Identified as a high-priority future development item.

CONCLUSION & SUMMARY

TrendScope — What Was Achieved

- ✓ 5-stage pipeline transforms any YouTube keyword into a full BI report in under 30 seconds
- ✓ Engagement rate (0.71 correlation) consistently outperforms raw view count (0.43) — validated across all 5 keywords — 65% improvement
- ✓ Power-law channel concentration confirmed in all 5 keywords (61–94% view concentration) — observed empirical pattern
- ✓ Market Gap Analysis identifies underserved content opportunities — the most actionable feature for real business decisions
- ✓ 16 interactive Plotly charts · 6-tab Streamlit app · 5-page Power BI dashboard — dual-interface delivery
- ✓ Live at grad-project-aeshaalami.streamlit.app — zero setup, free, publicly accessible worldwide

"The technology serves the business question — that is the fundamental principle of Business Intelligence."

TrendScope

Questions Welcome

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